

Simple Linear Regression

Jericho Lawson

MATH 240

Week 4, Lecture #1 — Ch. 7

Reminders

- Homework #3 has been assigned and is due **Friday** at 11:59pm
- Lab #2 during Friday's class

Introduction

We might be curious about whether a relationship exists between two different variables. Examples include:

- ① Does the amount of time spent in **daycare** before kindergarten affect the probability of showing **behavior problems** in elementary school?
- ② Is **average income** associated with **crime rate** in different districts?
- ③ Is **coffee consumption** in an office associated with the **growth of revenue**?

Regression can be used to examine the relationship between variables while holding other variables constant.

Types of Regression

Over the next few weeks, we will dive into the following regression frameworks:

- **Simple linear regression:** Modeling some explanatory variable X to attempt to explain response variable Y
- **Multiple linear regression:** Modeling multiple explanatory variables $X_1, X_2, \dots, X_p$ to attempt to explain response variable Y
- **Non-linear regression and interactions:** Modeling multiple explanatory variables $X_1, X_2, \dots, X_p$ in various forms (e.g. squared, interactions) to attempt to explain response variable Y
- **Logistic regression:** Modeling multiple explanatory variables $X_1, X_2, \dots, X_p$ to attempt to explain some binary response Y

- Relationship between two variables
 - ▶ Fitting line to data
 - ▶ Correlation coefficient
- Simple linear regression
 - ▶ Ideas
 - ▶ Use of residuals
 - ▶ Least squares
 - ▶ Interpretations
 - ▶ Strength
 - ▶ Use of categorical predictors

Objectives

- **Understand** how correlation works and what it translates to in linear regression
- **Explore** the reasoning behind the use of a least squares line
- **Calculate** β coefficients associated with linear regression
- **Discover** the limitations and strengths of simple linear regression

A Linear Relationship

All bivariate linear relationships can be summarized by one particular formula (which you've seen before):

$$y = mx + b$$

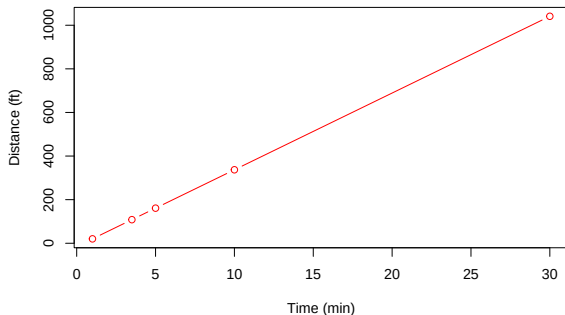
- m : slope
 - ▶ Represents the change in y for every one unit increase in x
- b : x -intercept
 - ▶ Represents value of y when $x = 0$
- y : response variable
- x : explanatory variable (i.e. covariate, predictor variable)

Example: Using Minutes to Determine Distance

Say we have the following equation that calculates how far we have to walk (in feet) to get to a destination y , using minutes walked for x :

$$y = 35.2x - 15$$

Distance Walked vs. Minutes Walked



Linear Regression

In reality, we will most likely not see a perfect linear relationship. We might collect data and get the following points:



This is where linear regression comes to play.

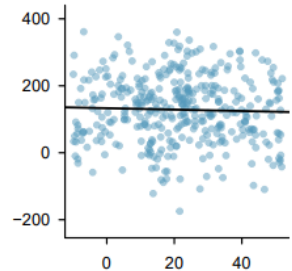
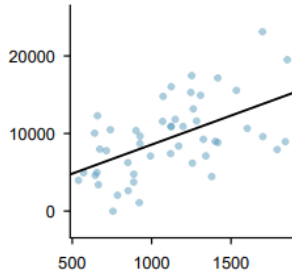
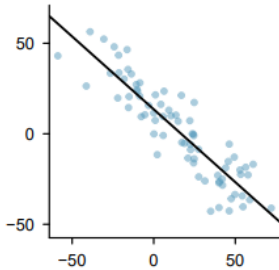
Linear Regression

Linear regression: Statistical method for fitting a line to data where the relationship between two variables (x and y) can be modeled by a straight line with some error:

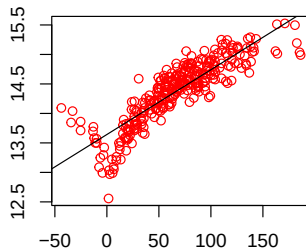
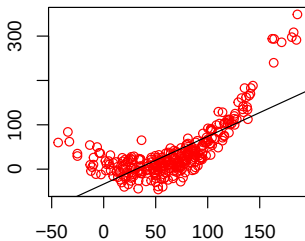
$$y = \beta_0 + \beta_1 x + \epsilon$$

- β_0 : intercept term
 - ▶ Represents value of y when $x = 0$
- β_1 : slope
 - ▶ Represents the change in y for every one unit increase in x
- ϵ : error
 - ▶ $\epsilon \sim N(0, \sigma^2)$

Examples



Not Examples



Correlation

Correlation: Describes the strength of the linear relationship between two variables

- Denoted as R
- Takes values between -1 and 1

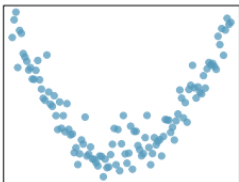
There are three characteristics to consider when looking at correlation:

- 1 **Direction:** from left to right, is it going up (positive) or down (negative)?
- 2 **Strength:** do the points signify more linearity (more extreme values)?
- 3 **Linearity:** do the points follow a linear trend?
 - ▶ If non-linear, values may not accurately represent relationship

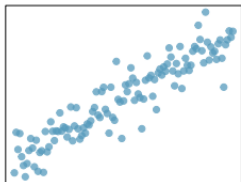
Activity: Identify the Relationships

Activity: Match each correlation to the corresponding scatter plot below. Explain the reasoning behind your matches.

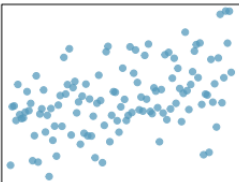
- (a) $R = -0.7$
- (b) $R = 0.45$
- (c) $R = 0.06$
- (d) $R = 0.92$



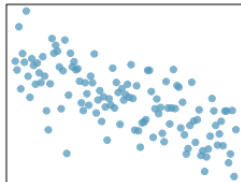
(1)



(2)



(3)



(4)

Correlation Coefficient Formulas

The correlation coefficient R can be calculated by the following formulas:

$$R = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$
$$= \frac{\sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{\left[\sum_{i=1}^n x_i^2 - \frac{1}{n} (\sum_{i=1}^n x_i)^2 \right] \left[\sum_{i=1}^n y_i^2 - \frac{1}{n} (\sum_{i=1}^n y_i)^2 \right]}}$$

- $\bar{x}$: mean of x
- $\bar{y}$: mean of y
- s_x : standard deviation of x
- s_y : standard deviation of y

Calculating Correlation Manually

Say we have the following data on the displacement (x) and MPG (y) of various cars from the late-20th century:

	1	2	3	4	5	6	7	8	9	10
x	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6
y	21.0	21.0	22.8	21.4	18.7	18.1	14.3	24.4	22.8	19.2

We can get the following values for the means and SDs: $\bar{x} = 208.61$, $\bar{y} = 20.37$, $s_x = 90.3739$, and $s_y = 2.9071$.

Then,

$$\begin{aligned} R &= \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right) \\ &= \frac{1}{10-1} \sum_{i=1}^{10} \left(\frac{x_i - 208.61}{90.3739} \right) \left(\frac{y_i - 20.37}{2.9071} \right) \\ &= -0.7739 \end{aligned}$$

Calculating Correlation Manually

Say we have the following values: $\sum_{i=1}^n x_i y_i = 40663.74$,
 $\sum_{i=1}^n x_i = 2086.1$, $\sum_{i=1}^n x_i^2 = 508688.3$, $\sum_{i=1}^n y_i = 203.7$, and
 $\sum_{i=1}^n y_i^2 = 4225.43$.

Then,

$$\begin{aligned} R &= \frac{\sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{\left[\sum_{i=1}^n x_i^2 - \frac{1}{n} (\sum_{i=1}^n x_i)^2\right] \left[\sum_{i=1}^n y_i^2 - \frac{1}{n} (\sum_{i=1}^n y_i)^2\right]}} \\ &= \frac{-1830.117}{2364.533} \\ &= -0.7739 \end{aligned}$$

Calculating Correlation through R

Say we have the following data on the displacement (x) and MPG (y) of various cars from the late-20th century:

	1	2	3	4	5	6	7	8	9	10
x	160.0	160.0	108.0	258.0	360.0	225.0	360.0	146.7	140.8	167.6
y	21.0	21.0	22.8	21.4	18.7	18.1	14.3	24.4	22.8	19.2

We can use the `cor()` function on the two variables to get the correlation coefficient:

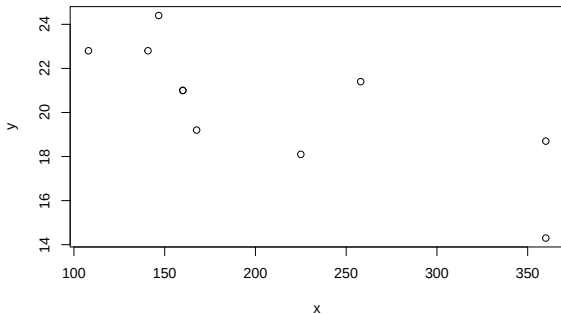
```
x = c(160.0, 160.0, 108.0, 258.0, 360.0, 225.0, 360.0, 146.7, 140.8, 167.6)
y = c(21.0, 21.0, 22.8, 21.4, 18.7, 18.1, 14.3, 24.4, 22.8, 19.2)
cor(x, y)
```

```
[1] -0.7739868
```

Understanding Correlation through R

We can also analyze the patterns of our two variables through a scatterplot:

```
plot(x, y)
```



Predictions and Residuals

Recall the setup for linear regression:

$$y = \beta_0 + \beta_1 x + \epsilon$$

Ultimately, we want to make predictions for y . To do this, we need to find estimates for β_0 and β_1 that will allow us to use the following formula:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Question: Where did the error go?

Residuals

- **Residual:** leftover variation in data after the model fit
 - ▶ Difference in response between the true value (y_i) and predicted value ($\hat{y}_i$) for a given x

$$e_i = y_i - \hat{y}_i$$

Observation is _____ the regression line.

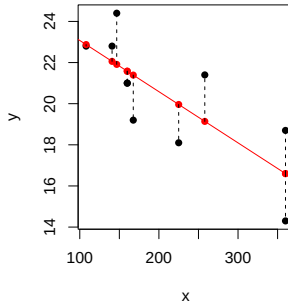
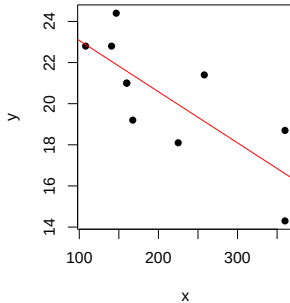
- **above**

- ▶ Residual will be positive (e.g. $e_i > 0$)
- ▶ $\hat{y}$ is underestimating y_i

- **below**

- ▶ Residual will be negative (e.g. $e_i < 0$)
- ▶ $\hat{y}$ is overestimating y_i

Residuals



$$\hat{y} = 25.5638 - 0.0249x$$

Analyzing Residuals

- We ultimately want to reduce the amount of error we have in our model
 - ▶ **Solution:** look at the residuals

We will need to minimize the sum of squared residuals:

$$\begin{aligned}\sum_{i=1}^n e_i^2 &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n \left[y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right]\end{aligned}$$

Thus, the values $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize this will make up our regression line, known as the **least squares line**.

Conditions to Note

When fitting the least squares line, we require the following conditions:

- 1 **Linearity**: data shows a linear trend
- 2 **Nearly normal residuals**: Residuals should follow approximately a normal distribution
- 3 **Constant variability**: variability around regression line should remain constant throughout all x
- 4 **Independent observations**: should not have some underlying structure (e.g. time series data)

We will explore these conditions in more detail when talking about diagnostics.

Formulas for Least Square Regression

We can estimate the intercept β_0 and the slope β_1 with the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ respectively, using the following formulas:

$$\begin{aligned}\hat{\beta}_1 &= \frac{s_y}{s_x} R \\ &= \frac{\sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sum_{i=1}^n x_i^2 - \frac{1}{n} (\sum_{i=1}^n x_i)^2}\end{aligned}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Activity

Suppose we look at the 10th observation in our cars dataset with a displacement (x) and MPG (y) of 167.6 and 19.2 respectively. Using the fitted line, find the following:

$$\hat{y} = 25.5638 - 0.0249x$$

- 1 Identify the estimated intercept and slope from the least squares line.
- 2 What is the true MPG for the displacement of 167.6?
- 3 What is the predicted MPG for the displacement of 167.6?
- 4 What is the residual?
- 5 Bonus: Find the correlation from this least squares line. (Use $s_x = 90.3739$ and $s_y = 2.907099$.)

For Wednesday!

- Homework #3 has been assigned and is due **Friday** at 11:59pm
- Read chapter 7 of IMS!